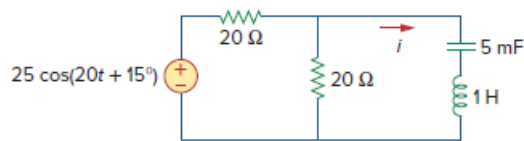


## Problem

Using source transformation, find  $i$  in the circuit of Fig. 10.94.

**Figure 10.94:**



## Step-by-step solution

## Step 1 of 5

Refer to Figure 10.94 in the circuit.

From the source, the angular frequency is,  $\omega = 20 \text{ rad/s}$ .

Convert the source voltage into phasor form.

$$25 \cos(20t + 15^\circ) = 25 \angle 15^\circ \text{ V}$$

Calculate the impedance of inductor.

$$\begin{aligned} Z_L &= j\omega L \\ &= j(20)(1) \\ &= j20 \Omega \end{aligned}$$

Calculate the impedance of capacitor.

$$\begin{aligned} Z_C &= \frac{1}{j\omega C} \\ &= \frac{1}{j(20)(5 \times 10^{-3})} \\ &= -j10 \Omega \end{aligned}$$

## Step 2 of 5

Draw the circuit diagram in Frequency domain.

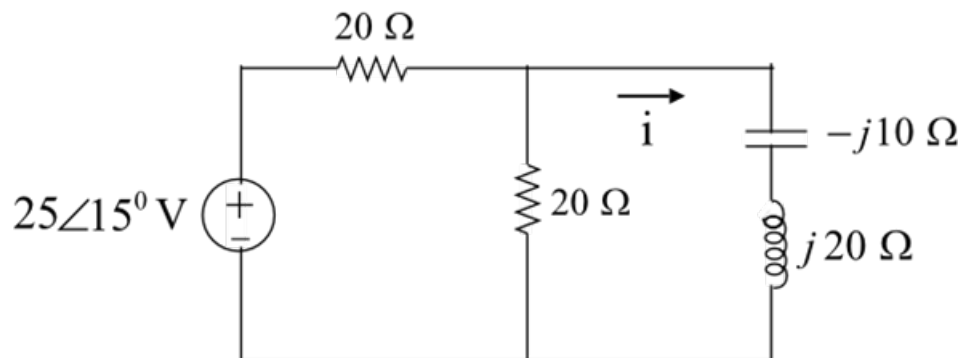


Figure 1

## Step 3 of 5

Convert the voltage source into current source using source transformation.

$$\begin{aligned} I_s &= \frac{25\angle 15^\circ}{20} \\ &= 1.25\angle 15^\circ \text{ A} \end{aligned}$$

Draw the modified circuit of Figure 1.

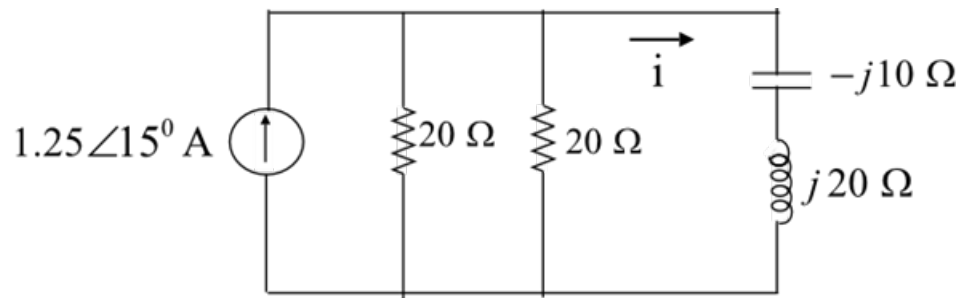


Figure 2

**Step 4** of 5

In Figure 2, the two resistors are connected in parallel. Calculate their equivalent resistance.

$$\begin{aligned} Z_1 &= \frac{20 \times 20}{20 + 20} \\ &= 10 \Omega \end{aligned}$$

Convert the current source to a voltage source using source transformation.

$$\begin{aligned} V_s &= I_s Z_1 \\ &= (1.25\angle 15^\circ)(10) \\ &= 12.5\angle 15^\circ \text{ V} \end{aligned}$$

Draw the modified circuit of Figure 2.

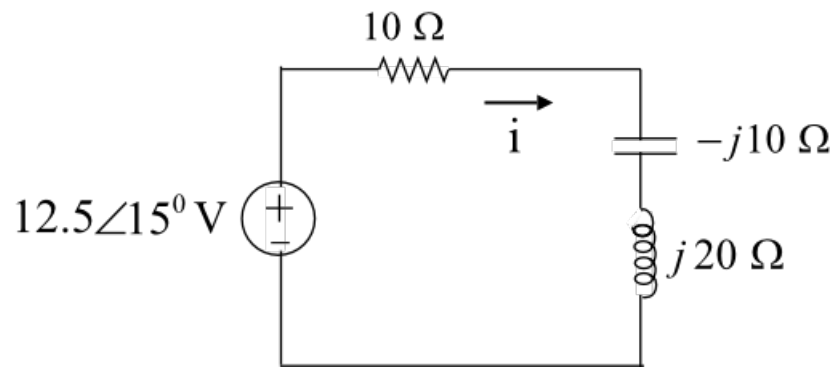


Figure 3

**Step 5** of 5

Calculate the current,  $I$ .

$$\begin{aligned} I &= \frac{12.5 \angle 15^\circ}{10 - j10 + j20} \\ &= \frac{12.5 \angle 15^\circ}{10 + j10} \\ &= \frac{12.5 \angle 15^\circ}{14.14 \angle 45^\circ} \\ &= 0.8839 \angle -30^\circ \text{ A} \end{aligned}$$

In the time domain, the value of current is,

$$i = 883.9 \cos(20t - 30^\circ) \text{ mA}$$

Therefore, the current,  $i$  is  $\boxed{883.9 \cos(20t - 30^\circ) \text{ mA}}$ .